EventOS

Developer Handbook

Derived from the repository at commit `eb056b3`

14 August 2026

Everything in this document was verified against the source.
Anything that could not be verified is labelled as such.

EventOS Developer Handbook

Version: derived from the repository at commit `f912143` (`origin/main`). See 22.3. **Generated:** 14 August 2026 **Scope:** the `wedding-planner-os` repository as it exists today

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How to read this handbook

Everything here was derived by reading the repository. Where a fact could not be established from the code, this document says **"Not implemented / not found in the current repository"** rather than filling the gap with a plausible guess.

Three labels appear throughout:

Label	Meaning
(verified)	Read directly out of the named file. Default — most statements are this.
(inference)	A reasonable reading of the code, not an explicit statement in it. Treat with care.
(not found)	Asked for, but absent from the repository.

File paths are relative to the repository root and were checked for existence. Function names are as they are exported.

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1. System Overview

1.1 What EventOS is

EventOS is a multi-tenant SaaS for **professional event planners**. The live product is Weddings; the marketing site (`src/app/(marketing)/page.tsx`) lists Corporate Events, Conferences, Birthdays and Galas as "Coming soon" — **none of those are implemented**.

The tenant is a **Studio**. A studio has planner users, weddings, guests, payments and support tickets. Every planner-facing query is scoped to one studio.

The product's distinguishing idea, stated on `src/app/(marketing)/weddings/page.tsx`: a guest does not get a shared wedding website, they get **their own** — showing only the events they are invited to, their own RSVP, and their own seat.

> **There is no couple-facing or guest-facing account.** Guests are identified by an unguessable invite code in a URL; they never register, log in or set a password. The only roles in the system are `ADMIN`, `PLANNER` and `MEMBER` (`prisma/schema.prisma`).

1.2 Current product capabilities

Verified by the existence of the corresponding routes and services:

Planner (`/studio/*`)

- Create, edit, duplicate, publish, unpublish and delete weddings — `src/server/services/weddings.ts`
- Choose one of six visual templates — `src/lib/themes.ts`
- Guest list with groups, CSV import and CSV export — `src/server/services/guests.ts`
- Send invitations in bulk, or resend to one guest — `src/server/services/invite-actions.ts`
- Multi-event schedule with per-guest visibility — `src/server/services/events.ts`
- RSVP tracking with meal and dietary notes — `src/server/services/rsvp.ts`
- Seating plan with tables and per-event seats — `src/server/services/seating.ts`
- Gift registry and cash funds — `src/server/services/registry.ts`
- Photo upload per slot (hero, couple, story, gallery) — `src/server/services/photos.ts`
- Studio branding: logo, brand colour, font — `src/server/services/branding.ts`
- Billing history and plan — `src/app/studio/billing/page.tsx`
- Help Center with 26 articles and support tickets — `src/lib/help.ts`, `src/server/services/support.ts`
- Must accept the current Terms of Service and Privacy Policy before any access — `src/server/services/legal.ts`

Platform admin (`/admin/*`)

- Create planner studios and issue invitations — `src/server/services/admin.ts`
- Suspend, reactivate, delete studios, reset planner passwords — same file

- Global and per-studio pricing — `src/server/services/pricing-admin.ts`
- Access requests queue — `src/server/services/access-requests.ts`
- Support ticket queue — `src/server/services/support.ts`
- Audit log viewer — `src/app/admin/activity/page.tsx`
- Payments list — `src/app/admin/payments/page.tsx`

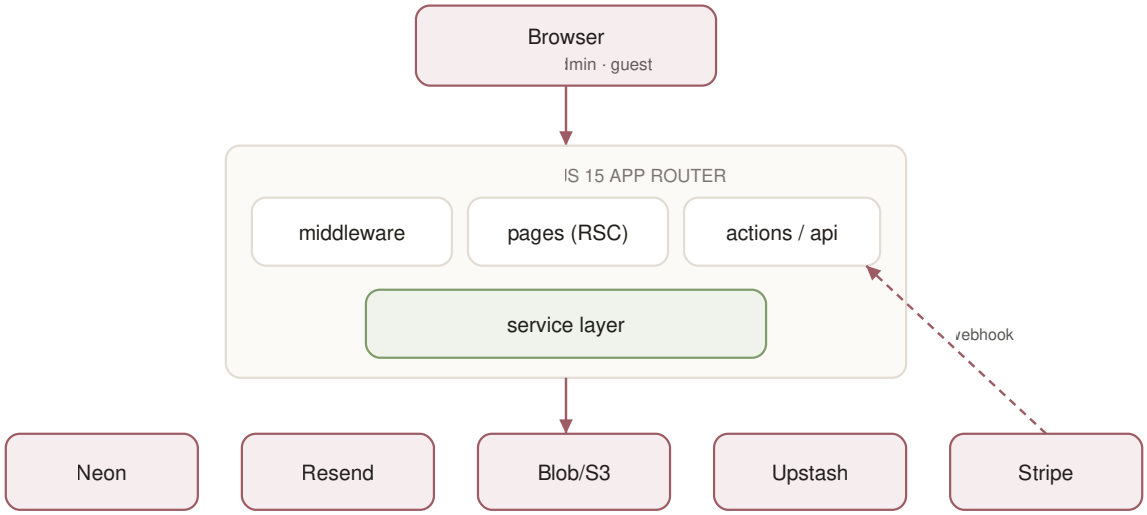
Guest (public, no account)

- Personal invitation portal at `/invite/[code]`
- Public wedding site at `/w/[slug]` (published weddings only)
- Registry at `/w/[slug]/registry`
- Calendar feed at `/calendar/[token]/[event]`

Public documents (no account)

- Terms of Service at `/terms` and Privacy Policy at `/privacy` — `src/app/(marketing)/terms/page.tsx`, `src/app/(marketing)/privacy/page.tsx`

1.3 High-level architecture



The shape to hold in your head: **pages and actions never touch the database directly** in the planner and admin trees — they call a service, and the service owns the tenant scoping. There are a small number of deliberate exceptions, noted in 4.5.

1.4 Technology stack

Read from `package.json`:

Layer	Choice	Version
Framework	Next.js (App Router)	`^15.1.4`
UI	React	`^19.0.0`
Language	TypeScript	`^5` (devDependency)
ORM	Prisma Client	`^6.2.0`
Database	PostgreSQL (Neon)	—
Auth	NextAuth / Auth.js	`^5.0.0-beta.25`
Passwords	bcryptjs	`^2.4.3`
Validation	Zod	`^3.24.1`
Email	Resend	`^4.0.1`
Payments	Stripe	`^17.5.0`
Blob storage	`@vercel/blob`	`^2.6.1`
S3 storage	`@aws-sdk/client-s3`	`^3.1100.0`
Images	sharp	`^0.35.3`
Ids	nanoid	`^5.0.9`
Smooth scroll	lenis	`^1.3.25`
Unit tests	Vitest	`^3` (dev)
E2E tests	Playwright	`1.62.1` (dev)

Not present: Remotion, Sentry, Redux, tRPC, tailwind. Styling is hand-written CSS in `src/app/globals.css`.

1.5 Major external services

Service	Used for	Configured by	Degrades to
Neon	PostgreSQL	`DATABASE_URL`, `DIRECT_URL`	Nothing — required
Vercel	Hosting, build, cron-free runtime	Platform	Nothing — required
Resend	All outbound email	`RESEND_API_KEY`, `EMAIL_FROM`	Emails recorded `SKIPPED`, app keeps working
Vercel Blob *or* **S3**	Photo and logo storage	`BLOB_READ_WRITE_TOKEN` or `S3_*`	Local filesystem driver in dev
Upstash Redis	Distributed rate limiting	`UPSTASH_REDIS_REST_URL/TOKEN`	In-process `Map` (see 15.7)
Stripe	Publishing payments, subscriptions	`STRIPE_SECRET_KEY`, `STRIPE_WEBHOOK_SECRET`	Publishing fails closed in production

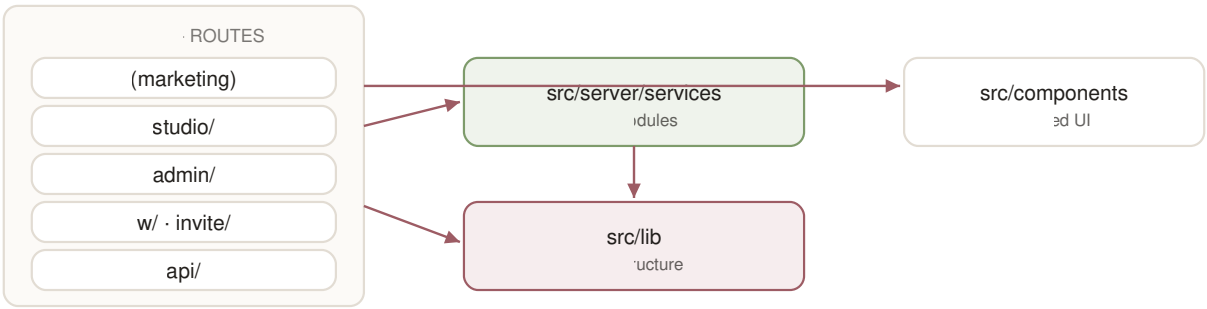
2. Repository Structure

2.1 Top level

```
wedding-planner-os/
├── prisma/              schema, migrations, seed, operational scripts
├── public/              static assets served verbatim (incl. the launch film)
├── scripts/             one-off developer scripts
├── src/                 all application code
├── tests/               unit tests (vitest) and e2e tests (playwright)
├── docs/                this handbook
├── .github/workflows/   CI
├── next.config.mjs      security headers, CSP, image config
├── playwright.config.ts E2E runner + E2E environment
├── vitest.config.ts     unit test runner + coverage thresholds
├── eslint.config.mjs    lint rules
└── package.json         scripts and dependencies
```

Markdown at the root is operational documentation, not code: `README.md`, `DEPLOYMENT.md`, `OPERATIONS.md`, `SECURITY.md`, `SECURITY_AUDIT.md`, `SECURITY-AUDIT-2026-08.md`, `AUDIT.md`, `EMAIL.md`, `SUPPORT.md`.

2.2 `src/` in detail



Folder	Purpose
`src/app/(marketing)/`	Public marketing site, login, forgot/reset password, request access. The `(marketing)` parentheses are a Next route group — they do not appear in URLs.
`src/app/studio/`	The planner application. Gated by `requireStudio()`.
`src/app/admin/`	Platform owner application. Gated by `requireAdmin()`.
`src/app/w/[slug]/`	Public wedding website. Published weddings only.
`src/app/invite/[code]/`	Personal guest portal. No account.
`src/app/calendar/[token]/[event]/`	`.ics` calendar feed.
`src/app/api/`	Route handlers: auth, health, ready, build, Stripe webhook.
`src/server/services/`	**All business logic.** 27 modules. Every one begins `import "server-only"`.
`src/lib/`	Infrastructure: database client, auth config, email, storage drivers, rate limiter, themes, validators, logging.
`src/components/`	Shared React components, both server and client.

2.3 Important individual files

File	Responsibility
`src/lib/auth.ts`	NextAuth configuration: credentials provider, JWT session, `__Host-` cookie, session revocation, timing equalisation. Exports `auth`, `signIn`, `signOut`, `handlers`, `SESSION_COOKIE`, `SessionUser`.
`src/lib/db.ts`	The Prisma client singleton. Five lines.
`src/server/services/context.ts`	**The authorization boundary.** `requireAdmin()`, `requireStudio()`, `ownWedding()`.
`src/lib/validators.ts`	Ten Zod schemas — every user-supplied object is parsed through one.
`src/lib/email.ts`	Resend integration, retry, redaction, `EmailLog` recording, `emailConfig()` diagnostics.
`src/lib/storage.ts`	Storage driver interface with three implementations (blob, s3, local).
`src/lib/storage-config.ts`	One shared predicate for "is storage configured", used by `env.ts`, `storage.ts` and `/api/ready`.
`src/lib/ratelimit.ts`	Upstash-backed limiter with in-process fallback.
`src/lib/lockout.ts`	Login throttle: per-account and per-IP counters, escalating delay.
`src/lib/env.ts`	Boot-time environment validation. Throws in production.
`src/lib/logger.ts`	Structured JSON logging with secret redaction.
`src/lib/themes.ts`	The six wedding templates and their palettes.
`src/lib/errors.ts`	`UserError` (safe to show) and `reportError()` (log real, return safe).
`src/lib/legal.ts`	`TERMS_VERSION`, `PRIVACY_VERSION`, and the review notice. Bumping a constant here re-gates every planner.
`src/middleware.ts`	Edge cookie-presence gate for `/studio` and `/admin`. **Not the security boundary.**
`next.config.mjs`	CSP, security headers, Server Action allowed origins, image remote patterns.
`prisma/schema.prisma`	25 models, 17 enums.
`prisma/seed.ts`	Development demo data.
`prisma/create-admin.ts`	Creates a real platform admin. Used on production and preview.

2.4 Generated and build artefacts

Never edit these; they are produced by tooling:

Path	Produced by
`.next/`	`next build`
`node_modules/`	`npm install`
`next-env.d.ts`	Next.js
`tsconfig.tsbuildinfo`	TypeScript incremental build
`coverage/`	`vitest run --coverage`
`test-results/`, `playwright-report/`	Playwright (both gitignored)
`src/lib/demo-photos.generated.ts`	`scripts/build-demo-photos.ts`
`prisma/migrations/*/migration.sql`	`prisma migrate dev` — **and must never be hand-edited once applied** (see 20.2)
`ph/`	Product Hunt marketing images. Not application code.

3. Frontend Architecture

3.1 App Router structure

Next.js 15 App Router. **Every page is a React Server Component by default.** Client components are the exception and are marked `"use client"`.

Route inventory, verified by `find src/app -name page.tsx`:

Public / marketing

Route	File
`/`	`src/app/(marketing)/page.tsx`
`/weddings`	`src/app/(marketing)/weddings/page.tsx`
`/login`	`src/app/(marketing)/login/page.tsx`
`/forgot-password`	`src/app/(marketing)/forgot-password/page.tsx`
`/reset-password/[token]`	`src/app/(marketing)/reset-password/[token]/page.tsx`
`/request-access`	`src/app/(marketing)/request-access/page.tsx`
`/terms`	`src/app/(marketing)/terms/page.tsx`
`/privacy`	`src/app/(marketing)/privacy/page.tsx`
`/accept-terms`	`src/app/accept-terms/page.tsx` — the legal gate
`/continue`	`src/app/continue/page.tsx`
`/dashboard`	`src/app/dashboard/page.tsx`

Guest-facing

Route	File
`/w/[slug]`	`src/app/w/[slug]/page.tsx`
`/w/[slug]/registry`	`src/app/w/[slug]/registry/page.tsx`
`/invite/[code]`	`src/app/invite/[code]/page.tsx`

Planner — all under `src/app/studio/`

`/studio`, `/studio/weddings`, `/studio/weddings/new`, `/studio/weddings/[id]`, `/studio/weddings/[id]/guests`, `/photos`, `/preview`, `/registry`, `/rsvps`, `/schedule`, `/seating`, `/studio/billing`, `/studio/settings`, `/studio/templates/[template]`, `/studio/help`, `/studio/help/[slug]`, `/studio/help/tickets`, `/studio/help/tickets/new`, `/studio/help/tickets/[id]`

Admin — all under `src/app/admin/`

`/admin`, `/admin/planners`, `/admin/planners/[id]`, `/admin/weddings`, `/admin/payments`, `/admin/settings`, `/admin/requests`, `/admin/activity`, `/admin/templates`, `/admin/support`, `/admin/support/[id]`

3.2 Layouts

Four layouts (`find src/app -name layout.tsx`):

Layout	Responsibility
`src/app/layout.tsx`	Root. Loads six Google fonts via `next/font` (Cormorant Garamond, Inter, Italiana, Pinyon Script, Playfair Display, Sacramento) and imports `globals.css`. Fonts are self-hosted with size-adjusted fallbacks to avoid layout shift.
`src/app/(marketing)/layout.tsx`	Marketing chrome — nav and footer.
`src/app/studio/layout.tsx`	Planner sidebar. Calls `requireStudio()`.
`src/app/admin/layout.tsx`	Admin sidebar. Calls `requireAdmin()`.

Calling the guard in the layout **and** in each page is deliberate: a layout is not a security boundary in the App Router, because a page can be rendered without its layout in some navigation cases. *(inference — the code does both consistently, and `src/app/admin/support/page.tsx` carries a comment to this effect.)*

3.3 Server vs client components

Client components — the complete list (18), from `grep -rl '"use client"' src/components`:

Component	Why it must be client
`src/components/InviteButtons.tsx`	`useActionState` for pending state on send/resend
`src/components/Wishlist.tsx`	`useActionState` for gift claiming
`src/components/RsvpForm.tsx`	Interactive form state
`src/components/Gallery.tsx`	Lightbox interaction
`src/components/Countdown.tsx`	Ticking timer
`src/components/Reveal.tsx`	Scroll-triggered animation
`src/components/SmoothScroll.tsx`	Lenis integration
`src/components/seating.tsx`	Drag-and-drop seating
`src/components/admin-dialogs.tsx`	Confirmation dialogs
`src/components/StudioBrandForm.tsx`	Live colour preview
`src/components/TimeZoneField.tsx`	Timezone picker
`src/components/BackLink.tsx`	Browser history navigation
`src/components/CustomDesignCard.tsx`	Request form state
`src/components/help/HelpSearch.tsx`	Help Center search
`src/components/marketing/AccessForm.tsx`	Access request form
`src/components/marketing/FilmPlayer.tsx`	Video playback
`src/components/marketing/SiteNav.tsx`	Marketing nav open/close
`src/components/AcceptLegalForm.tsx`	Checkbox state and pending state on the legal gate

Everything else is a Server Component. The pattern: **fetch on the server, pass plain data down, and make a component client only when it needs an event handler or a hook.**

3.4 State management

There is no state management library. No Redux, Zustand, Jotai or React Query.

State lives in four places:

1. **The database**, read fresh on each server render.
2. **The URL** — filters and editing state are query parameters (e.g. `/studio/weddings/[id]/guests?q=&group=&edit=`).
3. **`useActionState`** — per-form result and pending state in the client components above.
4. **A short-lived cookie** — the one-time password display on `/admin/planners` (`src/app/admin/planners/page.tsx`, `FLASH` cookie, 90 second `maxAge`).

Cache invalidation is `revalidatePath()` after mutations. There is no client cache to keep in sync.

3.5 Forms and Server Actions

Two shapes are used, and the choice matters.

A. Plain form action — for mutations where a full re-render or redirect is the right outcome:

```
async function remove(formData: FormData) {
  "use server";
  const { studioId } = await requireStudio();
  await deleteGuest(studioId, String(formData.get("guestId")));
  revalidatePath(`/studio/weddings/${...}/guests`);
}
```

B. `useActionState` with an outcome object — for mutations that can fail in an expected way and must not lose the page:

```
async function resend(_prev: InviteOutcome | null, formData: FormData): Promise<InviteOutcome> {
  "use server";
  const { studioId, user } = await requireStudio();
  return resendInvitationOutcome(studioId, String(formData.get("guestId")), user.name);
}
```

> **The rule that was learned the hard way:** an uncaught throw inside a Server Action is rendered by the error boundary, replacing the whole page. Any action that can raise a `UserError` — rate limits especially — must catch it and return a result. See `src/server/services/invite-actions.ts` and 16.

Every action re-derives its tenant with `requireStudio()` **inside the action**, never from the enclosing render's closure, because a Server Action is a separate request.

3.6 Navigation

`next/link` for internal navigation, `redirect()` from `next/navigation` inside actions.

`/dashboard` (`src/app/dashboard/page.tsx`) and `/continue` (`src/app/continue/page.tsx`) are role-routers: they read the session and forward an admin to `/admin` and a planner to `/studio`. They are explicitly documented in-file as *not* security boundaries.

3.7 Loading, error and 404 behaviour

File	Behaviour
`src/app/error.tsx`	Root error boundary. Client component. Shows a generic apology, a **Try again** button (`reset()`), a link to `/dashboard`, and `error.digest` as a support reference. **Renders nothing from the error object itself.**
`src/app/not-found.tsx`	Rendered by `notFound()`. Deliberately does not say "you don't have access", because 404-not-403 is how the app hides the existence of other tenants' data.

`loading.tsx` — not found in the current repository. There are no route-level loading files; pages render when their data resolves.

4. Backend Architecture

4.1 The service layer

`src/server/services/` holds 27 modules. Every one starts with `import "server-only"`, which makes importing it from a client bundle a build error.

Module	Responsibility
`access-requests.ts`	Public "request access" submissions and the admin queue
`admin.ts`	Create/suspend/delete planner studios, reset planner passwords
`audit.ts`	`logAudit()` — the single audit-write function
`billing.ts`	`startPublish()`, `completePublishFromStripe()`
`branding.ts`	Studio logo upload and removal
`calendar-feed.ts`	`.ics` generation
`context.ts`	**Authorization primitives**
`custom-design.ts`	Custom template requests
`events.ts`	Schedule events, and `personalEvents()` for per-guest visibility
`guests.ts`	Guest CRUD, CSV import, invitation sending
`idempotency.ts`	`runOnce()` — at-most-once side effects
`invite-actions.ts`	Outcome-returning wrappers for the two send paths
`legal.ts`	`hasAcceptedCurrentLegal()`, `outstandingLegal()`, `acceptCurrentLegal()`
`passwordReset.ts`	Token issue/consume, session revocation
`photos.ts`	Photo upload, replace, delete, reorder
`pricing.ts`	Price **reads** (no session dependency)
`pricing-admin.ts`	Price **writes** (admin only)
`registry.ts`	Gifts, cash funds, claiming
`rsvp.ts`	`submitRsvp()`
`seating.ts`	Tables and seats
`security-events.ts`	Pseudonymised security logging
`settings.ts`	`getSettings()` — the `PlatformSetting` singleton
`showcase.ts`	Public example weddings for the marketing page
`stripe-events.ts`	Webhook dispatch, idempotency
`subscriptions.ts`	Stripe customers, prices, subscription checkout, billing portal
`support.ts`	Support tickets, both planner and admin sides
`weddings.ts`	Wedding CRUD, duplicate, unpublish

Why `pricing.ts` and `pricing-admin.ts` are separate: reads happen on planner pages and inside the Stripe webhook, which has no session at all. Keeping `requireAdmin()` out of the read module stops the whole authentication stack being pulled in behind it.

4.2 Server Actions

Server Actions are defined inline in page files, not in a central actions directory. `next.config.mjs` pins their allowed origins:

```
allowedOrigins: [process.env.APP_URL, process.env.VERCEL_PROJECT_PRODUCTION_URL]
```

Body size limit is `4.5mb` (`next.config.mjs`), which bounds photo uploads.

4.3 API routes

Five route handlers under `src/app/api/`, plus two outside it.

Route	File	Auth	Purpose
`/api/auth/[...nextauth]`	`.../route.ts`	NextAuth	Sign-in/out endpoints
`/api/health`	`.../route.ts`	none	Liveness. Returns 200/503 with no body.
`/api/ready`	`.../route.ts`	none	Readiness: `{status, db, configured{...}, ms}`
`/api/build`	`.../route.ts`	**ADMIN**	Commit SHA/branch/message. 404s for everyone else.
`/api/webhooks/stripe`	`.../route.ts`	**Stripe signature**	Payment and subscription events
`/calendar/[token]/[event]`	`src/app/calendar/.../route.ts`	invite code or slug	`.ics` feed
`/studio/weddings/[id]/guests/export`	`src/app/studio/.../export/route.ts`	session + studio scope	Guest CSV

4.4 Validation

Ten Zod schemas in `src/lib/validators.ts`:

`zWedding`, `zGuest`, `zEvent`, `zRsvp`, `zGiftClaim`, `zRegistryItem`, `zStudioBranding`, `zAccessRequest`, `zTicket`, `zTicketReply`

Every free-text field is length-bounded — the security test suite asserts this, so a request cannot carry a megabyte of story text.

4.5 Database access

`src/lib/db.ts` — the entire file:

```ts
import { PrismaClient } from "@prisma/client";

const globalForPrisma = globalThis as unknown as { prisma?: PrismaClient };
export const prisma = globalForPrisma.prisma ?? new PrismaClient();
if (process.env.NODE_ENV !== "production") globalForPrisma.prisma = prisma;
```

The `globalThis` cache is only applied outside production, which is the correct pattern for serverless: it prevents hot-reload connection exhaustion in dev without keeping a stale client alive in a warm lambda.

`DATABASE_URL` points at Neon's **pooled** endpoint; `DIRECT_URL` at the direct endpoint, used by migrations, which need a real session a transaction pooler cannot give (`prisma/schema.prisma` datasource comment).

Pages that query Prisma directly rather than through a service — verified, and each re-derives the tenant first:

- `src/app/studio/weddings/[id]/guests/page.tsx` — the `editing` lookup, scoped `{ id, studioId, weddingId }`
- `src/app/admin/planners/[id]/page.tsx` — admin, already past `requireAdmin()`
- `src/app/admin/page.tsx`, `src/app/studio/billing/page.tsx` — aggregate reads

4.6 Authorization boundaries

`src/server/services/context.ts` is the whole boundary, and it is 29 lines:

```ts
export async function requireAdmin() {
  const session = await auth();
  const user = session?.user as SessionUser | undefined;
  if (!user || user.role !== "ADMIN") redirect("/login");
  return { user };
}

export async function requireStudio() {
  const session = await auth();
  const user = session?.user as SessionUser | undefined;
  if (!user || user.role !== "PLANNER" || !user.studioId) redirect("/login");
  const studio = await prisma.studio.findUnique({ where: { id: user.studioId } });
  if (!studio || studio.status === "SUSPENDED") redirect("/login");
  return { user, studio, studioId: studio.id };
}

export async function ownWedding(studioId: string, weddingId: string) {
  const wedding = await prisma.wedding.findFirst({ where: { id: weddingId, studioId } });
  if (!wedding) redirect("/studio/weddings");
  return wedding;
}
```

`requireStudio()` additionally gates on legal acceptance — see 7.8. The un-gated variant is `requireStudioSession()`, used by exactly one page.

Three things worth noticing:

1. `requireStudio()` re-reads the studio on every call, so **suspending a studio takes effect immediately** rather than at the next login.
2. `MEMBER` is in the `Role` enum but `requireStudio()` accepts only `PLANNER`. A MEMBER user cannot reach `/studio`. *(inference: the role appears reserved for future use; nothing in the repository grants it access.)*
3. `ownWedding()` **redirects rather than 403s**, so a foreign id and a non-existent id are indistinguishable from outside.

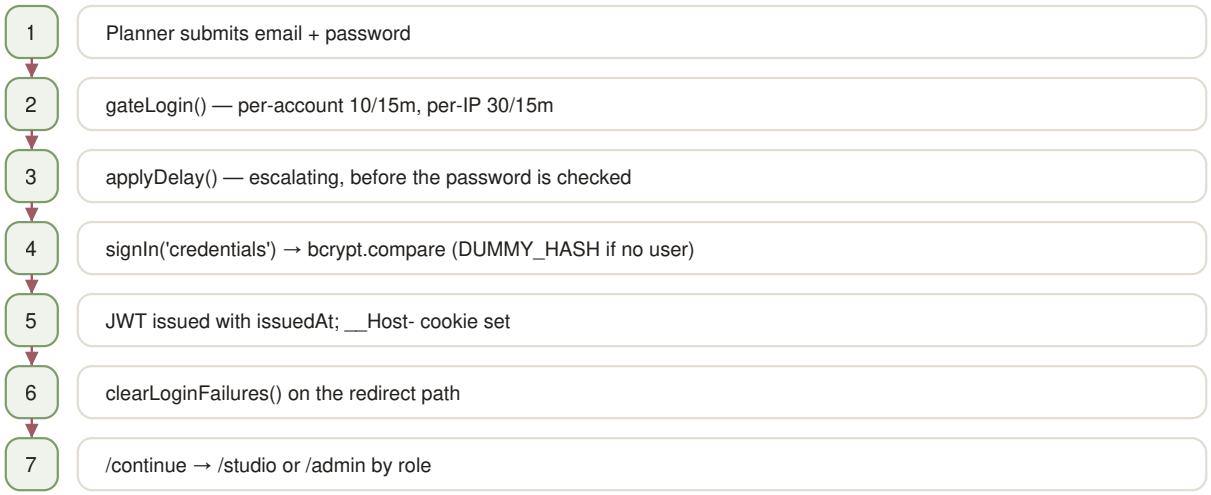
---

5. Data Flow

5.1 The general path

```
Browser
  → middleware.ts            (cookie present? else /login)
  → Server Component page    (requireStudio / requireAdmin)
  → Server Action            (re-derives tenant, parses input with Zod)
  → service                  (owns tenant scoping and business rules)
  → Prisma                   (query always carries studioId)
  → Postgres
  ← result
  ← revalidatePath / redirect / outcome object
  ← re-render
```

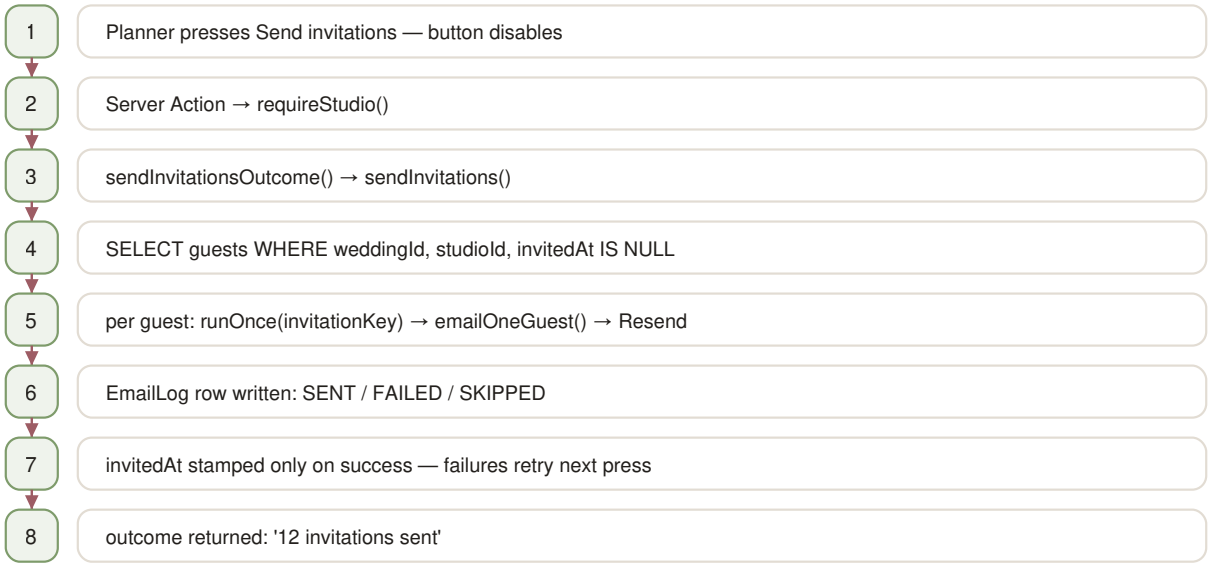
5.2 Authentication flow



Two details that exist for specific reasons:

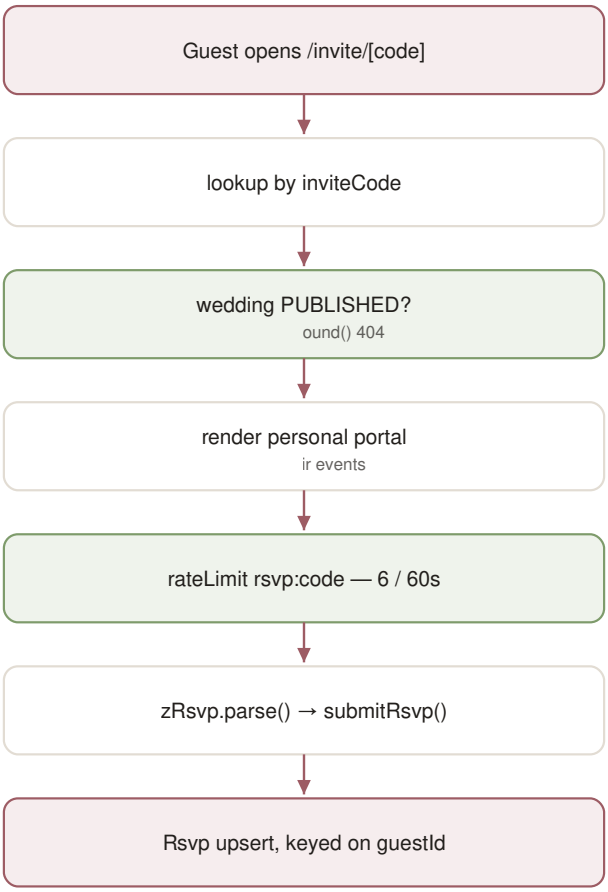
- **`DUMMY_HASH`** (`src/lib/auth.ts:42`) — when the email doesn't exist, bcrypt still runs against a dummy hash so the response time doesn't reveal whether an account exists.
- **`clearLoginFailures`** runs on the redirect path, detected by the NEXT_REDIRECT digest, because `signIn()` always leaves by throwing.

5.3 Invitation flow



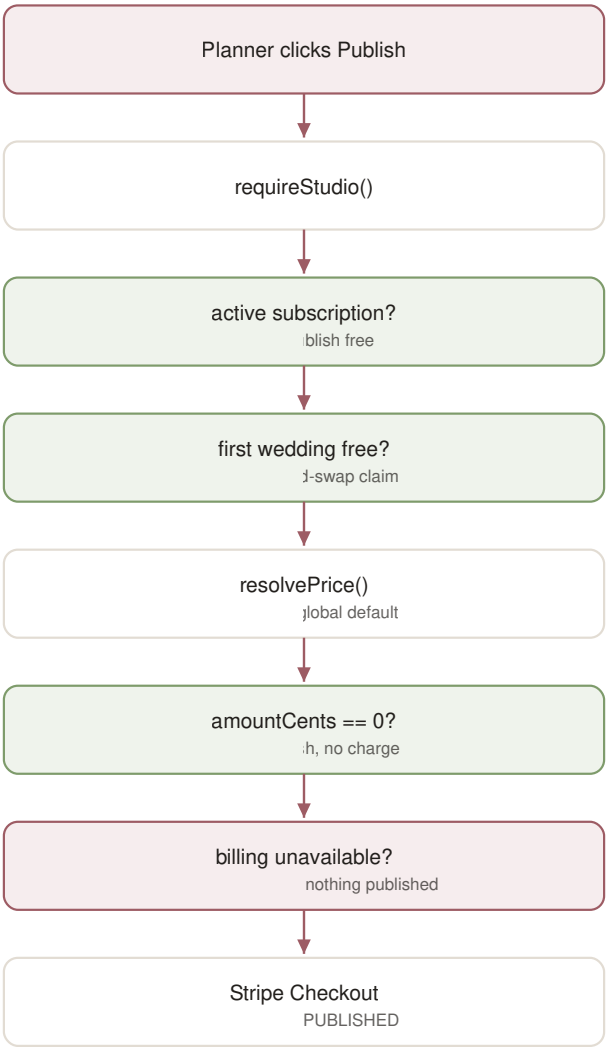
Partial success is structural: `invitedAt` is only stamped for guests the provider accepted, so failures stay un-invited and pressing the same button again retries exactly those.

5.4 RSVP flow



`Rsvp.guestId` is `@unique`, so a guest has at most one RSVP and re-submitting updates it.

5.5 Publishing flow



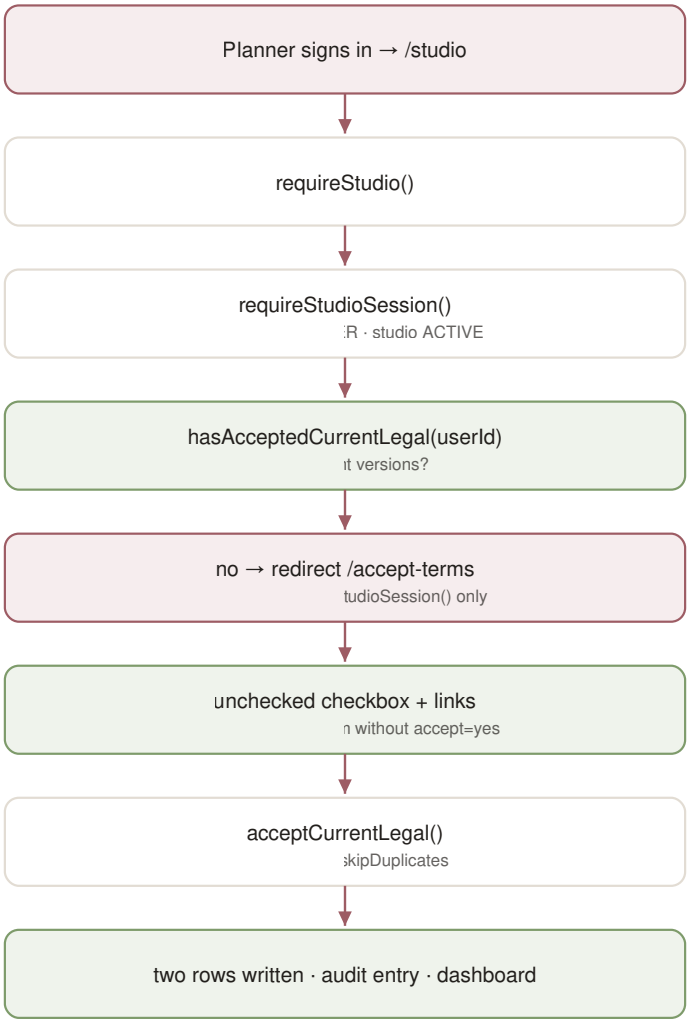
The free-wedding claim at **I** is a conditional `updateMany` rather than read-then-write, so two concurrent publishes cannot both take the free slot.

5.6 Email flow

```
service (e.g. guests.ts)
  → emails.<kind>()                        src/lib/email.ts
  → sendEmail()
      ├─ no RESEND_API_KEY  → record SKIPPED, return false   (no network)
      ├─ emailConfig() not ready → record SKIPPED, return false
      └─ attempt() with 3 tries, backoff [400ms, 1500ms]
            ├─ success → record SENT
            └─ failure → record FAILED (retryable errors retried)
  ← boolean
```

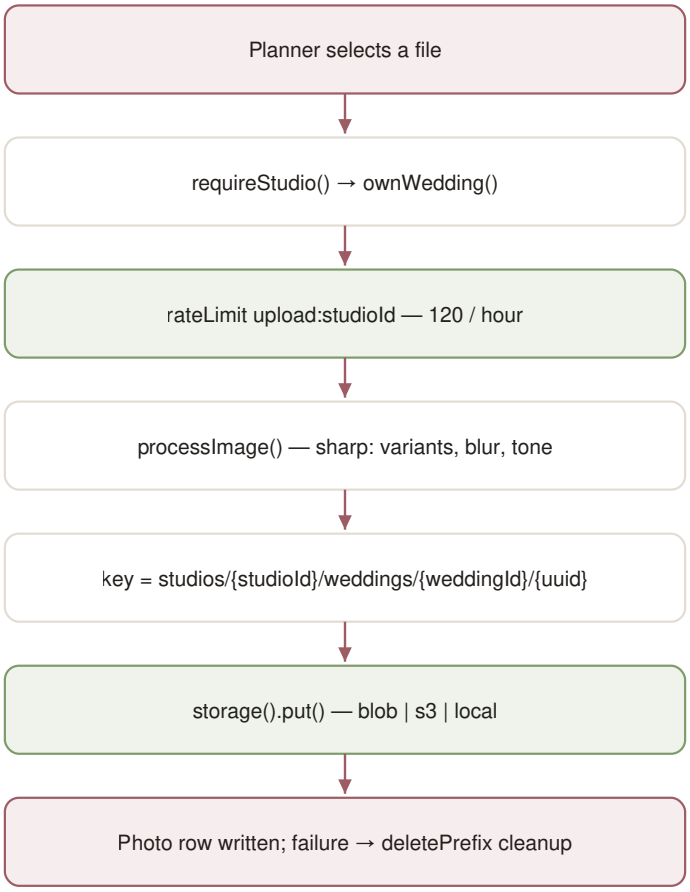
`sendEmail` never throws. Every outcome is a boolean plus an `EmailLog` row. This is why a bad address cannot crash a page — and why ignoring the return value silently hides bounces.

5.8 Legal acceptance flow



The gate is `requireStudio()`, which every planner page, the studio layout and every planner Server Action call. One surface cannot inherit it — `/studio/weddings/[id]/guests/export` is a route handler and route handlers do not run layouts — so it carries the same check inline and returns **403** rather than a redirect, because redirecting a fetch would hand the caller a CSV-shaped file full of HTML.

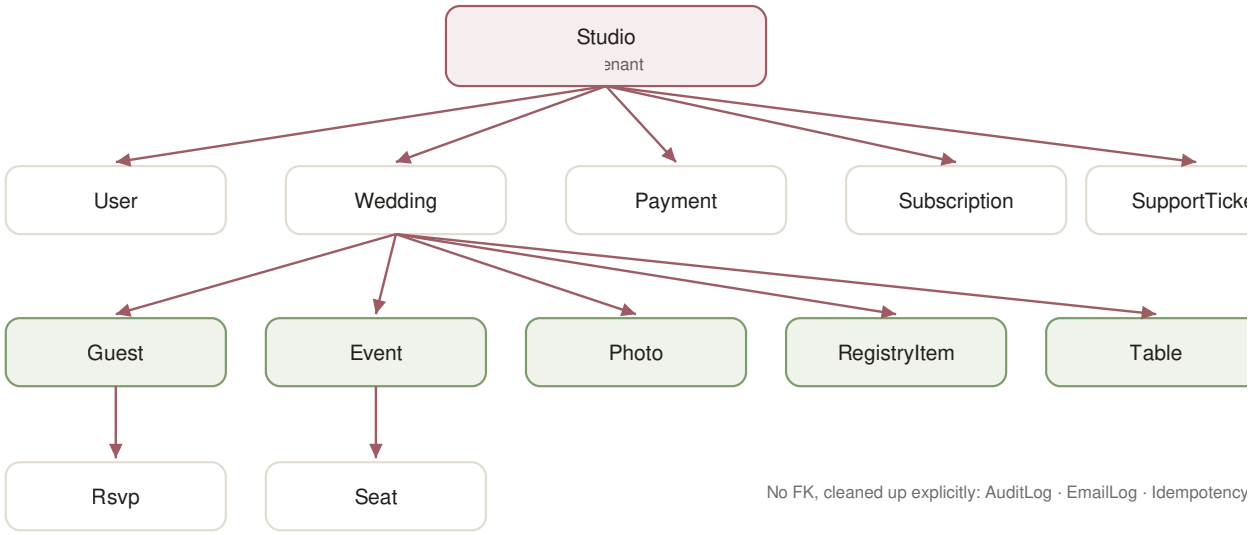
5.7 Upload flow



6. Database

PostgreSQL via Prisma. `prisma/schema.prisma` defines **25 models** and **17 enums**.

6.1 Entity relationship diagram



Not shown, because they intentionally have **no foreign keys**: `AuditLog`, `EmailLog`, `IdempotencyKey` carry a plain `studioId` column so a log outlives the record it describes. `ProcessedWebhookEvent`, `PlatformSetting` and `AccessRequest` stand alone.

6.2 Models

Model	Key fields	Notes
`User`	`email @unique`, `passwordHash`, `role`, `studioId?`, `sessionsValidFrom?`	`sessionsValidFrom` is the JWT revocation cutoff
`PasswordResetToken`	`tokenHash @unique`, `expiresAt`, `usedAt?`	Token stored hashed, never raw
`Studio`	`slug @unique`, `stripeCustomerId @unique`, `freeWeddingUsed`, branding fields	The tenant
`Wedding`	`slug @unique`, `template`, `status`, `publishedAt?`	
`Guest`	`inviteCode @unique`, `email?`, `groups`, `invitedAt?`	`invitedAt` gates bulk sending
`Table`	`shape`, `eventId`, `weddingId`	Seating
`Seat`	`@@unique([guestId, eventId])`	One seat per guest per event
`Event`	`startsAt`, visibility fields	The schedule
`Rsvp`	`guestId @unique`, `status`, `meal`, `dietary`, `notes`	
`RegistryItem`	`purchasedBy?`	Claiming uses a conditional update
`CashFund`		
`Faq`		
`Payment`	`stripeSessionId @unique`, `stripePaymentIntentId @unique`, `stripeInvoiceId @unique`, `pricePlanId?`	
`PricePlan`	`activeKey @unique`, `amountCents`, `studioId?`, `archivedAt?`	Immutable price versions
`Subscription`	`studioId @unique`, `stripeSubscriptionId @unique`, `pricePlanId`	Mirrored from Stripe
`ProcessedWebhookEvent`	`stripeEventId @unique`	Webhook idempotency
`PlatformSetting`	`id Int @default(1)`	Singleton
`AuditLog`	`actorType`, `action`, `studioId?`	No FK — survives deletion by design
`Photo`	`basePath`, `variants Json`, `blurData`, `tone Json?`	
`EmailLog`	`toEmail`, `kind`, `status`, `studioId?`	No FK
`AccessRequest`	`status`	Public sign-up requests
`IdempotencyKey`	`key @unique`, `completedAt?`, `expiresAt`	Backs `runOnce()`
`SupportTicket`	`studioId`, `userId`, `status`, `priority`, `category`	
`TicketMessage`	`authorType`	
`LegalAcceptance`		Append-only consent record

Model	Key fields	Notes
	`userId`, `document`, `version`, `acceptedAt`, `@@unique([userId, document, version])`	

6.3 Enums

`Role` (ADMIN, PLANNER, MEMBER) · `StudioStatus` (ACTIVE, SUSPENDED) · `WeddingStatus` (DRAFT, PUBLISHED, ARCHIVED) · `RsvpStatus` (ACCEPTED, DECLINED, MAYBE) · `PaymentStatus` (PENDING, PAID, FAILED, REFUNDED) · `PricePlanKind` (PER_WEDDING, MONTHLY, YEARLY) · `SubscriptionStatus` (8 values mirroring Stripe exactly) · `TemplateKey` (6 templates) · `EmailStatus` (SENT, FAILED, SKIPPED) · `PhotoSlot` (HERO, COUPLE, STORY, GALLERY) · `TableShape` (ROUND, RECTANGLE, SQUARE, HEAD) · `AccessRequestStatus` (NEW, INVITED, DECLINED) · `TicketStatus` (5) · `TicketPriority` · `TicketAuthor` · `TicketCategory` · `LegalDocument` (TERMS, PRIVACY)

> **`ARCHIVED` exists in `WeddingStatus` but no code path sets it.** *(verified by grep — no service assigns `status: "ARCHIVED"`.)*

6.4 Cascades and delete behaviour

20 `onDelete: Cascade` relations. Deleting a `Studio` removes its users, weddings, guests, events, photos, payments, tickets and price overrides.

Two `onDelete: Restrict` — both protecting `PricePlan`:

- `Payment.pricePlan` (schema line 461)
- `Subscription.pricePlan` (line 542)

A price that has billed someone cannot be deleted out from under its receipt.

Three tables deliberately have no FK and are cleaned up explicitly in `deletePlanner()` (`src/server/services/admin.ts`): `EmailLog`, `AuditLog`, `IdempotencyKey`. Uploaded blobs are removed in the same function via `storage().deletePrefix("studios/{id}/")`, best-effort and after the database work.

6.5 Indexes and constraints worth knowing

54 `@unique` / `@@unique` / `@@index` declarations. The load-bearing ones:

Constraint	Why it exists
`PricePlan.activeKey @unique`	Holds `"<kind>:<studioId>"` or `"<kind>:GLOBAL"` while live, NULL when archived. Because Postgres treats NULLs in a unique index as distinct, unlimited archived rows coexist while two rows can never both be current for one scope.
`IdempotencyKey.key @unique`	Makes `runOnce()` race-safe — the loser of an insert gets P2002 and returns the existing outcome.
`ProcessedWebhookEvent.stripeEventId @unique`	Stripe replays; the index makes a replay a no-op.
`Rsvp.guestId @unique`	One RSVP per guest.
`Seat @@unique([guestId, eventId])`	A guest cannot be seated twice at one event.
`Guest.inviteCode @unique`	The guest's identity.
`LegalAcceptance @@unique([userId, document, version])`	Makes acceptance idempotent — a double-click or retry cannot write a second row — and is what lets `createMany({ skipDuplicates: true })` be the whole write path, with no read-then-check.

6.6 Migration history

18 migrations, in order. Each name states its purpose:

Migration	Purpose
`20260729203159_init`	Initial schema
`20260730185726_audit_email_log_password_reset`	AuditLog, EmailLog, PasswordResetToken
`20260731120000_photos`	Photo model
`20260731210000_travel_fields_photo_tone`	Travel info, photo tone analysis
`20260801090000_seating`	Tables and seats
`20260803160000_seating_per_event`	Seating became per-event
`20260803170000_access_request`	Public access requests
`20260804090000_calendar_and_maps`	Calendar feed and map fields
`20260804140000_registry_purchases`	Gift claiming
`20260805090000_midnight_bloom_template`	Template added
`20260806090000_pacific_linen_template`	Template added
`20260807090000_velvet_botanical_template`	Template added
`20260808090000_idempotency_keys`	`runOnce()` support
`20260809090000_studio_branding`	Logo, brand colour, font
`20260810090000_session_revocation`	`User.sessionsValidFrom`
`20260812090000_support_tickets`	Support system
`20260812120000_price_plans_and_subscriptions`	Versioned pricing + Stripe subscriptions. **The only migration that seeds data** — it inserts the three default price plans.
`20260814120000_legal_acceptance`	`LegalAcceptance` table and `LegalDocument` enum. **Deliberately no backfill** — every existing planner is left without a row so the gate stops them and they accept explicitly. Seeding agreement on someone's behalf would record consent that was never given.

Migrations run automatically: `"build": "prisma generate && prisma migrate deploy && next build"`.

> **This means `npm run build` migrates whatever `DATABASE_URL` is in scope.** It is the single most dangerous command in the repository. Use `npx next build` when you only want to compile.

6.7 Seed data

`prisma/seed.ts` (`npm run db:seed`) creates development demo data: admin `owner@weddingos.app`, studio "Prestige Weddings", planner `sarah@prestigeweddings.com`, one wedding `sarah-and-james`, with the shared password `password123`.

Never run this against production or any internet-reachable database. The credentials are public knowledge in this repository. `prisma/create-admin.ts` (`npm run db:create-admin -- <email>`

`<name> <password>`) is the production-safe alternative: it creates exactly one ADMIN, requires a 12+ character password, and is idempotent.

6.8 Production vs Preview vs E2E databases

Three Neon branches, each with its own connection string:

Environment	Branch	Set via
Production	`production`	Vercel Production env vars
Preview	`preview`	Vercel Preview env vars
E2E	`e2e`, database `eventos_e2e`	`E2E_DATABASE_URL` / `E2E_DIRECT_URL`

`playwright.config.ts` assigns `E2E_DATABASE_URL` onto `process.env.DATABASE_URL` before workers spawn, so the test workers and the test server cannot disagree about which database they are using.

`tests/e2e/seed.ts` contains `assertTestDatabase()`, which refuses to seed unless the URL matches `/test|e2e|localhost|127\.0\.0\.1/i`. That is why the e2e database is named `eventos_e2e` — so the guard passes because it is true, not because it was loosened.

---

7. Authentication and Authorization

7.1 Login

`src/app/(marketing)/login/page.tsx` → `gateLogin()` → `signIn("credentials")`.

`src/lib/lockout.ts` defines the throttle:

Counter	Limit	Window
Per account (`login:acct:<pseudonymised>`)	`ACCOUNT_HARD` = 10	15 min
Per IP (`login:ip:<ip>`)	`IP_HARD` = 30	15 min

A soft threshold introduces an escalating delay before the password is checked, so the cost of guessing rises. The account key is a **pseudonymised** hash of the email (`src/server/services/security-events.ts`, `pseudonymise()`), so neither the limiter keys nor the security log become a list of everyone who has tried to sign in.

7.2 Sessions and cookies

From `src/lib/auth.ts`:

Setting	Value
Strategy	`jwt` (no session table)
`maxAge`	12 hours (`SESSION_MAX_AGE_S`)
`updateAge`	15 minutes (`SESSION_UPDATE_AGE_S`)
Cookie name	`__Host-authjs.session-token` in production
Flags	`httpOnly`, `sameSite: "lax"`, `secure: true`, `path: "/"`
Claim refresh	`CLAIM_REFRESH_MS` = 60 s

`__Host-` is the strictest cookie prefix the platform offers: it requires Secure, a path of `/`, and forbids a Domain attribute, which stops a subdomain from setting it.

Revocation: JWTs are stateless, so `User.sessionsValidFrom` is the cutoff. Every token carries its issue time and the `jwt` callback drops any token older than the column. This is what makes a password reset actually sign other sessions out.

7.3 Password hashing and reset

Hashing: bcrypt, cost factor 12, everywhere (`bcrypt.hash(password, 12)`).

Reset, in `src/server/services/passwordReset.ts`:

1. `issueResetToken()` — `crypto.randomBytes(32).toString("hex")`, stored as `sha256(token)` in `PasswordResetToken.tokenHash`. The raw token exists only in the email.
2. `resetPassword()` — rejects if missing, already used, or expired.
3. Consumption is a **compare-and-swap**: `updateMany({ where: { id, usedAt: null } })`, so two concurrent uses of one link cannot both succeed.
4. All of the user's other outstanding tokens are marked used.
5. `revokeSessionsOp()` runs in the same transaction.

Invitation tokens use `INVITE_TOKEN_TTL_MS` = 7 days.

Both `/forgot-password` and the reset page are rate-limited by IP (`forgot:` 5/60s, `reset:` 10/10min), and `/forgot-password` returns the same response whether or not the account exists.

7.4 Roles and permissions

Capability	ADMIN	PLANNER	MEMBER
`/admin/*`	✓	✗	✗
`/studio/*`	✗	✓	✗
Create studios, reset planner passwords	✓	✗	✗
Set prices	✗	✓	✗
Manage own weddings/guests	✗	✓	✗

An ADMIN has no `studioId` and is redirected out of `/studio` exactly as a stranger would be. This is deliberate and the E2E helper `assertSessionUsable` accounts for it.

7.5 Tenant isolation

The rule: **every planner-facing query carries `studioId`, and the check and the write are one statement.**

```
// correct — check and write are atomic
await prisma.supportTicket.updateMany({ where: { id, studioId }, data: {...} });

// wrong — a window exists between the two
const t = await prisma.supportTicket.findUnique({ where: { id } });
if (t.studioId !== studioId) throw ...;
```

`tests/tenancy.test.ts` asserts this at the layer that enforces it, using a fake Prisma client so that what is tested is *the query the service builds*.

7.6 IDOR protections

Surface	Protection
`/studio/weddings/[id]/*`	`ownWedding()` — redirect, not 403
Support tickets	`getMyTicket(studioId, id)` uses `findFirst({ where: { id, studioId } })`
Guest CSV export	Route handler re-derives studio from session, then `findFirst({ id, studioId })`
`/w/[slug]`	Requires `status: "PUBLISHED"`
`/invite/[code]`	Requires the wedding be PUBLISHED; 404 otherwise
Calendar feed	404 for unknown token, never 403
Photos, seating, registry	`studioId` denormalised onto the row for O(1) checks

7.7 Middleware vs server-side authorization

`src/middleware.ts` checks only that **a session cookie exists** on `/admin/:path*` and `/studio/:path*`. It deliberately does not decode the JWT, because importing `lib/auth` would pull Prisma and bcrypt into the edge runtime where neither can run.

> The middleware is **fast-path UX, not the security boundary.** `requireAdmin()` and `requireStudio()` are. A forged cookie gets past middleware and is rejected a moment later by the server-side check.

The `next` parameter is set from `req.nextUrl.pathname` only — never the full URL — so the login page cannot be turned into an open redirect.

7.8 The Terms and Privacy gate

A planner has no access to their account until they have affirmatively agreed to the versions of the Terms of Service and Privacy Policy currently in force.

Where it lives. `requireStudio()` in `src/server/services/context.ts`. That function is the chokepoint: the studio layout calls it, all 21 studio pages call it, and every planner Server Action calls it again on its own request rather than trusting the render that produced the form. One check therefore covers page loads, direct URL navigation and mutations.

Not middleware. `src/middleware.ts` runs on the edge where Prisma cannot, and only checks that a cookie exists. A gate there would be a redirect, not a boundary.

The split. `requireStudioSession()` is everything `requireStudio()` used to be — session, role, active studio — with no legal check. Exactly one page uses it: `/accept-terms`, which needs a session to know whose consent it is recording but obviously cannot require the consent it exists to collect.

Why `/accept-terms` sits outside `/studio`. `src/app/studio/layout.tsx` calls `requireStudio()`, so a page under that layout would be redirected to itself forever. Keeping the screen at the root means the layout stays fully gated rather than being weakened to accommodate one page.

The one surface that cannot inherit it. `src/app/studio/weddings/[id]/guests/export/route.ts` is a route handler, and Next does not run layouts for route handlers. It carries the check inline and returns 403. Without that line, the one thing an un-accepted planner could still download would be every guest's name and email address.

Versioning. `TERMS_VERSION` and `PRIVACY_VERSION` in `src/lib/legal.ts`. The check is equality against those constants, so bumping one re-gates every planner on their next request — no migration, no backfill, no second switch. Old rows stay, so "what had this person agreed to on the day they did X" has an answer.

What is not collected. No IP address, no user-agent. Deliberate: they are the obvious additions for evidentiary weight and also personal data the product has no other reason to hold.

Who is exempt. Admins. `requireAdmin()` has no legal gate — this is an agreement between EventOS and the planners who use it. Guests are never asked; they hold no account and are not a party to it.

The documents. `/terms` and `/privacy` are public, because a planner must be able to read them from a screen that has not yet let them in, and a prospective customer should be able to read them before signing up. Both carry a visible notice that they are a draft pending review by a qualified attorney.

---

8. Wedding Lifecycle

State	Meaning	Set by
`DRAFT`	Private to the studio. Guest URLs 404.	`createWedding()`
`PUBLISHED`	Live. `publishedAt` stamped.	`startPublish()` / `completePublishFromStripe()`
`ARCHIVED`	**In the enum, never set by any code path.**	—

Preview is not a status — it is the route `/studio/weddings/[id]/preview`, which renders the guest site for the planner while the wedding is still a draft (`src/components/PreviewBar.tsx` shows the banner).

`duplicateWedding()` copies a wedding as a new DRAFT — useful for a studio running similar events.

9. Guests and Invitations

9.1 Guest creation

Three routes in, all through `src/server/services/guests.ts`:

- `addGuest(studioId, weddingId, input)` — single, validated by `zGuest`
- `importGuests(studioId, weddingId, actor, csv)` — bulk CSV; returns `{imported, skipped}`, skipping lines with a missing name or malformed email
- Guests can also arrive with a wedding via `duplicateWedding()`

Each guest gets an `inviteCode` from `inviteCode()` in `src/lib/utils.ts`:

```
const codeAlphabet = customAlphabet("ABCDEFGHJKMNPQRSTUVWXYZ23456789", 10);
```

10 characters from a 31-character alphabet ≈ 49.5 bits of entropy. The alphabet omits `I`, `L`, `O`, `0` and `1` — characters a guest could misread off a printed card. `tests/security.test.ts` asserts both the entropy and the omissions.

9.2 Personalization

`personalEvents()` in `src/server/services/events.ts` returns only the events a given guest is invited to, based on their `groups`. The same wedding therefore renders differently for each guest — this is the product's core claim.

9.3 Sending

Path	Function	Bounded by
Bulk	`sendInvitations(studioId, weddingId, actor)`	`invitedAt: null` — cannot re-send
Single	`resendInvitation(studioId, guestId, actor)`	`rateLimit("resend:<guestId>", 3, 1h)`

Bulk sending is paced by `INVITE_SEND_INTERVAL_MS` between messages — a burst of identical messages from a new domain is what spam filters watch for.

Each bulk send goes through `sendInvitationOnce()` → `runOnce()` with `invitationKey(weddingId, guestId, inviteCode)` and a 6-hour TTL. The invite code is part of the key deliberately: regenerating a guest's code is how a planner deliberately re-issues an invitation, and that should be allowed through immediately.

Resend deliberately does **not** use `runOnce` — re-sending is the whole point of the button. The rate limit is what bounds it.

9.4 Rate limits — complete list

Key	Limit	Window	Where
`resend:<guestId>`	3	1 hour	`guests.ts`
`upload:<studioId>`	120	1 hour	`photos.ts`
`logo:<studioId>`	20	1 hour	`branding.ts`
`gift:<weddingId>`	30	1 hour	`registry.ts`
`rsvp:<inviteCode>`	6	60 s	`invite/[code]/page.tsx`
`support:<studioId>`	10	1 hour	`support.ts`
`custom-design:<studioId>`	5	24 hours	`custom-design.ts`
`access:<ip>`	3	1 hour	`access-requests.ts`
`forgot:<ip>`	5	60 s	`forgot-password/page.tsx`
`reset:<ip>`	10	10 min	`reset-password/[token]/page.tsx`
`login:acct` / `login:ip`	10 / 30	15 min	`lockout.ts`

10. Templates

10.1 The six templates

Defined in `src/lib/themes.ts`, enumerated in `TemplateKey`:

`BLUSH_ROMANCE` · `MODERN_SAGE` · `CLASSIC_ELEGANCE` · `MIDNIGHT_BLOOM` · `PACIFIC_LINEN` · `VELVET_BOTANICAL`

`FALLBACK_TEMPLATE = "BLUSH_ROMANCE"`. `themeFor()` falls back rather than throwing when a row names a template this build has never heard of — a database outlives a deployment, and `tests/security.test.ts` covers exactly this case.

10.2 Architecture

```
Wedding.template (TemplateKey enum)
  → themeFor(key)          src/lib/themes.ts  → Theme object
  → themeVars(theme)       → CSS custom properties
  → WeddingSite.tsx        src/components/WeddingSite.tsx
  → rendered guest site
```

`isDarkTheme` drives contrast decisions. `src/lib/photo-tone.ts` and `src/components/SitePhoto.tsx` adapt image treatment per template.

10.3 Adding a new template

Every step verified against the three template migrations that already exist (`20260805090000_midnight_bloom_template` and siblings):

1. **`prisma/schema.prisma`** — add the value to `enum TemplateKey`.
2. **Create a migration** — `npm run db:migrate` generates the `ALTER TYPE ... ADD VALUE`.
3. **`src/lib/themes.ts`** — add the entry to `THEMES` with its palette and fonts.
4. **`src/lib/utils.ts`** — add to `TEMPLATES` (the display registry used by pickers).
5. **`src/app/studio/templates/[template]/page.tsx`** — verify the preview renders.
6. **Fonts** — if a new face is needed, add it to `src/app/layout.tsx` via `next/font`.
7. **Tests** — `tests/security.test.ts` asserts every registered template resolves to a complete palette; it will fail if the entry is incomplete.

> **Common mistake:** adding the enum value without the `THEMES` entry. `themeFor()` silently falls back to `BLUSH_ROMANCE`, so the template *appears* to work and quietly renders the wrong design.

11. Email

11.1 Architecture

`src/lib/email.ts` is the only module that talks to Resend. `src/lib/email-render.ts` renders the HTML and text bodies.

```
const resend = process.env.RESEND_API_KEY ? new Resend(process.env.RESEND_API_KEY) : null;
const SANDBOX_FROM = "onboarding@resend.dev";
const FROM = process.env.EMAIL_FROM ?? `EventOS <${SANDBOX_FROM}>`;
```

11.2 Email kinds

The `EmailKind` union — 10 values:

`GUEST_INVITATION`, `RSVP_CONFIRMATION`, `PLANNER_INVITE`, `CUSTOM_DESIGN_REQUEST`, `PAYMENT_RECEIPT`, `PASSWORD_RESET`, `ACCESS_REQUEST`, `ACCESS_REQUEST_ACK`, `SUPPORT_TICKET_OPENED`, `SUPPORT_TICKET_REPLY`

Note `EmailKind` is a **TypeScript union, not a database enum** — `EmailLog.kind` is a plain `String`.

11.3 Sender configuration

`EMAIL_FROM` accepts both `Name <address>` and a bare address. It is used two ways:

1. **As the From header** — passed through verbatim.
2. **Extracted to a bare address** in four places, all using the identical expression: `ts from.match(/<([^>]+)>/)?.[1] ?? (from.includes("@") ? from.trim() : null)` `src/lib/email.ts:96` (`emailConfig().fromAddress`), `src/lib/email.ts:296` (`PLATFORM_INBOX`), `src/server/services/access-requests.ts:30`, `src/server/services/support.ts:72`.

`emailConfig()` reports problems: missing key, still using the sandbox sender in production, or a consumer mailbox domain (gmail/outlook/…) whose DMARC policy would reject the mail. `npm run email:check` prints this.

11.4 Failure handling

Three attempts with backoff `[400ms, 1500ms]`, retrying only transient failures. Every attempt lands in `EmailLog` as `SENT`, `FAILED` or `SKIPPED` with the provider error. **`sendEmail` never throws.**

11.5 E2E behaviour

`playwright.config.ts` sets `RESEND_API_KEY: ""` in `webServer.env`, so the suite makes no network calls to Resend. This works because `@next/env` only adopts a `.env` value when the key is `undefined` in the parent environment — an empty string wins.

---

12. Storage and Uploads

12.1 Drivers

`src/lib/storage.ts` defines one interface and three implementations:

```ts
export interface StorageDriver {
  readonly name: "blob" | "s3" | "local";
  put(key: string, body: Buffer, contentType: string): Promise<StoredObject>;
  deletePrefix(prefix: string): Promise<void>;
  publicUrl(key: string): string;
}
```

Driver	Selected when	Notes
`blobDriver()`	`BLOB_READ_WRITE_TOKEN` or `BLOB_STORE_ID` present	`access: "public"`, `addRandomSuffix: false`
`s3Driver()`	all of `S3_BUCKET`, `S3_ACCESS_KEY_ID`, `S3_SECRET_ACCESS_KEY`	also reads `S3_ENDPOINT`, `S3_REGION`, `S3_PUBLIC_URL`
`localDriver()`	neither configured	writes to `public/uploads`, dev only

`src/lib/storage-config.ts` holds the single shared predicate (`storageConfigured()`) used by `env.ts`, `storage.ts` and `/api/ready`, so the three cannot drift apart.

12.2 Keys

```
studios/{studioId}/weddings/{weddingId}/{randomUUID()}    photos
studios/{studioId}/brand/{randomUUID()}                   logos
```

`randomUUID()` makes keys unguessable — important because blob objects are public-read. One prefix per studio is what makes `deletePrefix("studios/{id}/")` a complete cleanup.

12.3 Processing and deletion

`src/lib/images.ts` (sharp): `processImage()`, `processLogo()`, `asVariants()`, `srcSet()`, `fallbackSrc()`. Multiple sizes plus a `blurData` placeholder are produced per upload; `src/lib/photo-tone.ts` measures luminance/saturation so the renderer can adapt.

Deletion happens in four places, all `deletePrefix`: photo delete, photo replace, logo replace, logo removal — plus studio deletion in `admin.ts`.

Security note: stored objects are publicly readable by URL. Access control is entirely key unguessability. There is no signed-URL mechanism. *(verified — no signing code exists.)*

---

13. Environment Variables

Every variable below is referenced somewhere in `src/`, `prisma/`, `scripts/` or a config file. **No values appear in this document.**

13.1 Required in production

Variable	Purpose	Notes
`DATABASE_URL`	Pooled Postgres connection	Server-only
`DIRECT_URL`	Direct Postgres connection	Used by migrations
`AUTH_SECRET`	JWT signing	Must be ≥32 chars; `env.ts` rejects known placeholder prefixes
`APP_URL`	Absolute base for links in emails	`env.ts` rejects localhost in production
`UPSTASH_REDIS_REST_URL`	Distributed rate limiting	Required in production by `env.ts`
`UPSTASH_REDIS_REST_TOKEN`	ditto	ditto

`src/lib/env.ts` throws at boot (via `src/instrumentation.ts` → `register()`) if any are missing in production, failing the deploy rather than serving a broken app.

13.2 Expected — feature degrades without them

Variable	Purpose	Without it
`RESEND_API_KEY`	Email sending	Emails recorded `SKIPPED`
`EMAIL_FROM`	From header	Falls back to the Resend sandbox sender
`STRIPE_SECRET_KEY`	Payments	Publishing fails closed in production unless price is $0
`STRIPE_WEBHOOK_SECRET`	Webhook verification	Webhook returns 400
`BLOB_READ_WRITE_TOKEN` *or* `S3_*`	Photo storage	Local driver (dev only)

13.3 Optional

Variable	Purpose
`EMAIL_REPLY_TO`	Reply-to on platform email
`ACCESS_REQUEST_TO`	Where access requests go; falls back to the address in `EMAIL_FROM`
`SUPPORT_TICKET_TO`	Where ticket notifications go
`SHOWCASE_WEDDING_SLUGS`	Which weddings appear on the marketing page
`AUTH_TRUST_HOST`	NextAuth behind a proxy
`BLOB_STORE_ID`	Alternative blob detection
`S3_ENDPOINT`, `S3_REGION`, `S3_PUBLIC_URL`	Non-AWS S3 targets

13.4 Platform-injected (do not set by hand)

`VERCEL_ENV`, `VERCEL_GIT_COMMIT_SHA`, `VERCEL_GIT_COMMIT_REF`, `VERCEL_GIT_COMMIT_MESSAGE`, `VERCEL_DEPLOYMENT_ID`, `VERCEL_PROJECT_PRODUCTION_URL`, `NODE_ENV`, `NEXT_RUNTIME`, `CI`

13.5 Test-only

`E2E_DATABASE_URL`, `E2E_DIRECT_URL`, `E2E_BASE_URL`, `E2E_PORT`, `E2E_APP_URL` — read only by `playwright.config.ts`.

13.6 Client exposure

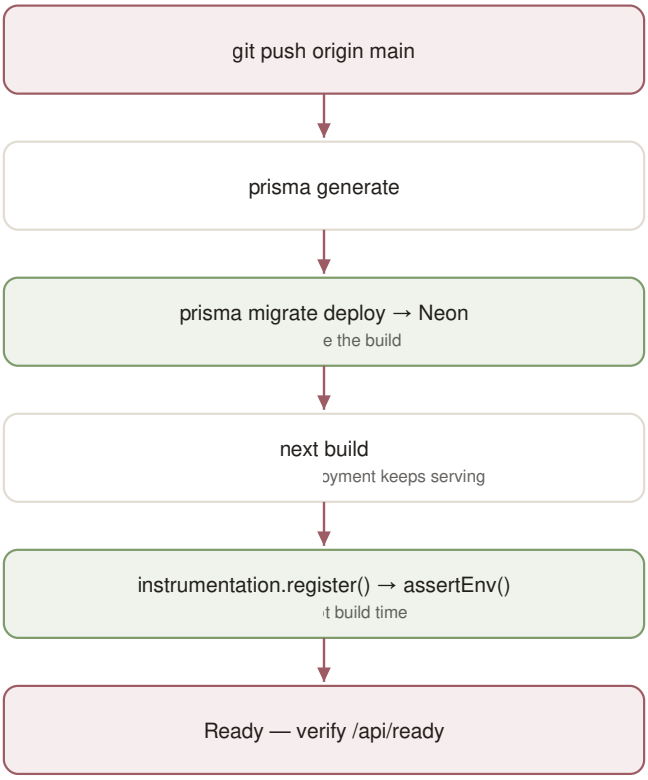
> **There are no `NEXT_PUBLIC_*` variables in this repository.** *(verified by grep.)* Every variable above is server-only and none is bundled into client JavaScript. Any future variable prefixed `NEXT_PUBLIC_` is embedded in the browser bundle and must never hold a secret.

14. Deployment

14.1 Local development

```
npm install
cp .env.example .env      # then fill in your own values
npm run db:migrate        # or db:deploy
npm run db:seed           # demo data — never against a shared database
npm run dev
```

14.2 Vercel



> **The failure mode to know about:** `assertEnv()` runs at *runtime* boot, not build time. A missing production variable therefore produces a green build and a completely broken deployment. Always check `/api/ready` after deploying.

14.3 Preview

Preview uses its own Neon branch. The branch must be **empty**, not a schema-only copy — a schema-only branch has the tables but an empty `_prisma_migrations`, so `migrate deploy` replays `init` and fails with `P3018: type "Role" already exists`.

`scripts/reset-preview-db.ts` empties a preview branch safely. It refuses to run against a database containing any application rows, which makes pointing it at production a no-op rather than a catastrophe.

14.4 E2E environment

`npx playwright test` builds for production and starts a server on port 3100. `playwright.config.ts` supplies an environment that satisfies the production guard without weakening it: an `https://e2e.invalid` `APP_URL`, unreachable Upstash on `127.0.0.1:1` (so `consume()` takes its documented fallback), and an empty `RESEND_API_KEY`.

14.5 Verification after deploy

```
curl -s https://www.youreventos.com/api/ready
```

Expect `status: "ready"` and the `configured` block. Then check `/api/health` returns 200 and sign in.

14.6 Rollback

Vercel keeps previous deployments; promoting an earlier one is the fastest rollback for **application** code.

> **Migrations do not roll back.** `prisma migrate deploy` has no down-migration. Rolling the code back while the schema has moved forward can produce a mismatch. This is why additive migrations matter — see 20.2.

---

15. Security Architecture

A full audit lives in `SECURITY_AUDIT.md`. This section describes the mechanisms.

Control	Implementation
Authentication	NextAuth credentials, bcrypt cost 12, timing-equalised via `DUMMY_HASH`
Session	JWT, `__Host-` prefix, 12 h, revocable via `sessionsValidFrom`
Authorization	`requireAdmin()` / `requireStudio()` at every page and service
Tenant isolation	`studioId` in every planner query; check+write in one statement
IDOR	404/redirect rather than 403 everywhere
Password reset	32-byte token, stored hashed, single-use CAS, revokes sessions
Rate limiting	11 distinct limits; Upstash-backed, in-process fallback
Webhook	`stripe.webhooks.constructEvent` over the **raw body**, plus event-id idempotency
Legal gate	`requireStudio()` blocks every planner surface until both current documents are accepted; the CSV route repeats it inline
CSRF	Server Actions with `allowedOrigins` pinned; `sameSite=lax` cookies
Headers	CSP, HSTS (2 years, preload), X-Frame-Options, X-Content-Type-Options, Referrer-Policy, COOP, CORP, X-DNS-Prefetch-Control
CSP	`default-src 'self'`, `object-src 'none'`, `frame-ancestors 'none'`, `base-uri 'self'`, `form-action 'self'`
Secrets	None in the repository; `src/lib/logger.ts` redacts connection strings, API keys, JWTs and `whsec_` values before logging
CSV injection	Guest export prefixes `=`, `+`, `-`, `@` cells with an apostrophe

Known CSP weakness: `script-src` includes `'unsafe-inline'`, which defeats most of CSP's XSS value. The actual defences are React's escaping and the absence of any `dangerouslySetInnerHTML` in the codebase *(verified by grep — zero occurrences)*.

Rate-limit fallback: `consume()` degrades to the in-process limiter when Redis is unreachable, and logs `ratelimit.redis_unavailable`. This is deliberate — refusing every request when the limiter is down converts a dependency blip into a total outage.

Client IP: taken from `x-forwarded-for`. This is safe on Vercel specifically, because Vercel overwrites that header and does not forward external IPs. If a proxy is ever placed in front of Vercel, switch to `x-vercel-forwarded-for`.

16. Error Handling

Two error classes, from `src/lib/errors.ts`:

Class	Meaning	Shown to user
`UserError(message, code?)`	Expected, actionable	Yes — the message was written for them
Anything else	A bug or infrastructure fault	No — one sentence plus a reference id

`reportError(scope, err, fallback)` draws the line: a `UserError` is logged at `warn` and its own message returned; anything else is logged at `error` with its stack and reduced to `"<fallback> (reference abc123)"`.

The pattern that matters:

```
// in a Server Action
const outcome = await someServiceOutcome(...);   // returns {ok, message}
// NOT: await someService(...)  ← an uncaught UserError replaces the page
```

`src/app/error.tsx` renders nothing from the error object — in production Next replaces server-side error messages with a generic string before they cross to the client, leaving only `digest`, so there is nothing safe *or* useful to display beyond the reference.

17. Logging and Observability

17.1 Structured logging

`src/lib/logger.ts` — one JSON line per event to stdout, which Vercel forwards to any attached drain. `log.debug/info/warn/error(event, fields)`.

Redaction is the reason it exists. Errors near a database call routinely carry a connection string; `console.error(err)` would print it into a searchable, months-retained log. The module strips connection strings, API keys, JWTs, `whsec_` values and guest email addresses, and survives cyclic objects.

17.2 Audit log

`logAudit()` in `src/server/services/audit.ts` writes `AuditLog` rows with `actorType` (ADMIN / PLANNER / GUEST / SYSTEM), `action`, `studioId?`, `targetId?`. Visible at `/admin/activity` and per-studio on `/admin/planners/[id]`.

17.3 Email log

Every send attempt lands in `EmailLog` with status and provider error. This is the first place to look when a planner says an invitation never arrived.

17.4 Health and readiness

Endpoint	Answers
`/api/health`	"is this process alive" — 200/503, no body
`/api/ready`	"is this instance configured to do its job" — `{status, db, configured{database, auth, email, storage, payments, distributedRateLimit}, ms}`

`status` is `ready` when everything is configured, `degraded` when email or storage is missing, `unavailable` (503) when the database or auth is broken.

`sweepIdempotencyKeys()` is called from the health probe rather than a cron — the table is small and the delete is indexed.

17.5 What is intentionally not monitored

- **No APM or error tracker.** No Sentry, no Datadog. Deliberate at current scale.
- **No cron jobs.** There is no `vercel.json`; nothing is scheduled.
- **No uptime monitoring in-repo** — `OPERATIONS.md` records that Better Stack polls `/api/health` externally.

18. Testing

18.1 Suites

Suite	Runner	Count	Command
Unit / service	Vitest	214 across 13 files	`npm test`
E2E security	Playwright	29	`npx playwright test`

18.2 Unit test files

File	Covers
`tests/tenancy.test.ts`	Every service query carries `studioId`
`tests/security.test.ts`	Logger redaction, invite-code entropy, slugify, validators, rate limiter, themes, **security headers**
`tests/races.test.ts`	Free-wedding claim and gift claim under concurrency
`tests/support.test.ts`	Ticket authorization both directions
`tests/pricing.test.ts`	Price resolution and versioning
`tests/billing-subscriptions.test.ts`	Publish pricing, duplicate charges, webhook idempotency
`tests/production-hardening.test.ts`	Fail-closed billing, complete deletion, live-deployment predicate
`tests/invite-actions.test.ts`	Resend failure paths, message hygiene
`tests/reliability.test.ts`	Retry, degradation
`tests/branding.test.ts`	Logo handling
`tests/storage-config.test.ts`	The shared storage predicate
`tests/audit-fixes.test.ts`	Regressions from earlier audits
`tests/legal.test.ts`	The Terms/Privacy gate: version equality, idempotent writes, admin exemption

The testing philosophy worth preserving: these use a **fake Prisma client**, not a database, because what is under test is *the query the service builds*. A real database with one studio in it would let an unscoped query pass by accident.

18.3 E2E

`tests/e2e/` — `authorization.spec.ts` (13), `abuse.spec.ts` (5), `headers.spec.ts` (3, one parameterised over three paths so it reports 5), `legal.spec.ts` (6).

They run against a **production build** deliberately: the `__Host-` cookie prefix, the security headers and the boot-time env guard only exist there.

`tests/e2e/helpers.ts` contains two functions worth understanding before editing anything:

- `assertSessionUsable()` — every sign-in ends by fetching a page only that account may see and requiring 200. Without it, four tenancy tests once passed while proving nothing, because "redirected to /login" satisfied their assertion.
- `expectBouncedFromStudioResource()` — explicitly **rejects** a `/login` redirect, because an unauthenticated caller would otherwise satisfy a tenancy assertion.

18.4 Running E2E

```
export E2E_DATABASE_URL='<neon e2e branch, pooled>'
export E2E_DIRECT_URL='<neon e2e branch, direct>'
npx playwright test
```

Coverage thresholds are per-module in `vitest.config.ts`, and every number was measured before it was written down.

---

19. Debugging Runbook

Login fails

1. `/api/ready` — is `auth: true`?
2. `?error=rate` in the URL means the lockout tripped: 10 per account or 30 per IP in 15 min. Wait or check Upstash.
3. `?error=1` is a bad credential.
4. Session set but immediately signed out → `AUTH_SECRET` changed, or `User.sessionsValidFrom` is in the future.
5. Cookie not stored → the site must be HTTPS; `__Host-` requires Secure.

Password reset fails

1. Check `EmailLog` for a `PASSWORD_RESET` row. `SKIPPED` means email isn't configured.
2. Link expired (`expiresAt`) or already used (`usedAt`) — both refuse by design.
3. `reset:<ip>` limit is 10 per 10 minutes.

Invitation not received

1. **`EmailLog` first** — filter by `toEmail`. `SENT` means Resend accepted it; the problem is downstream.
2. `SKIPPED` → `RESEND_API_KEY` missing, or `emailConfig()` reported a problem.
3. `FAILED` → the provider error is in the row.
4. Verify the domain is verified in Resend and `EMAIL_FROM` is on it.
5. `npm run email:check` prints the whole configuration.

Resend button fails

1. Three per guest per hour. The fourth returns a message beside the button — this is correct behaviour, not a bug.
2. If the **page** breaks rather than the button, a Server Action is throwing: it must return an outcome (see `src/server/services/invite-actions.ts`).

RSVP fails

1. Wedding must be `PUBLISHED` — otherwise `/invite/[code]` 404s.
2. `rsvp:<code>` allows 6 per minute.
3. `zRsvp` rejects a status outside `ACCEPTED | DECLINED | MAYBE`.

Wedding won't publish

1. **Most likely:** Stripe is not configured and the price is non-zero → `BILLING_UNAVAILABLE`. Either configure Stripe or set the per-wedding price to `$0` in `/admin/settings`.
2. `No active PER_WEDDING price is configured` → the price-plan seed did not run.
3. Already-published weddings return `{ok: true}` and do nothing.

Planner stuck on /accept-terms

1. They must tick the box — the server ignores a submission without `accept=yes`.
2. If they accept and are sent straight back, check `LegalAcceptance` for rows matching **both** current versions in `src/lib/legal.ts`. A row for an older version does not count, by design.
3. A planner who reaches the gate unexpectedly usually means a version constant was bumped.

Migration fails

Error	Meaning	Action
`P3018` + `already exists`	Schema present, `_prisma_migrations` empty (schema-only branch)	Empty the branch — do **not** `migrate resolve`
`P1001`	Cannot reach the server	Check the connection string host
`P1013`	Malformed URL	Usually shell quoting; single-quote the whole value
`P2002`	Unique violation	Two rows claiming one constraint

Storage / upload fails

1. `/api/ready` → `storage`.
2. `120` uploads per studio per hour.
3. Body limit is `4.5mb` (`next.config.mjs`).
4. Driver in use is logged at boot: `[storage] driver=…`.

Vercel deployment fails

1. Build log: `prisma migrate deploy` runs before `next build`.
2. Build green but every route 500s → `assertEnv()` threw at boot. The message names the missing variables.

`/api/health` or `/api/ready` failing

`health` 503 → the process cannot serve. `ready` 503 → database unreachable or auth unconfigured. `degraded` → email or storage missing; the app still works.

Rate-limit issues

`distributedRateLimit: false` in `/api/ready` means Upstash is unset and limits are per-instance. Look for `ratelimit.redis_unavailable` in the logs.

---

20. Safe Development Guidelines

20.1 Modifying the database

1. Edit `prisma/schema.prisma`.
2. `npm run db:migrate` against a **local or branch** database.
3. Review the generated SQL by hand.
4. Run `npm test`.
5. Commit schema and migration together.

20.2 Migration rules

> **Never edit a migration that has been applied anywhere.** Prisma stores a checksum; a changed file makes the next `migrate deploy` fail on every environment that already ran it.
>
> **Prefer additive migrations.** There is no down-migration. An additive change lets you roll application code back without a schema mismatch.

Adding a required column to a populated table needs three deploys: add nullable → backfill → make required.

20.3 Modifying services

- Keep `import "server-only"` at the top.
- Take `studioId` as a parameter; never read it from input.
- Keep check and write in one statement (`updateMany` with the tenant in `WHERE`).
- Throw `UserError` for expected failures; let real faults propagate.
- Add the query-shape test to `tests/tenancy.test.ts`.

20.4 Adding an API route

Decide the auth model first — session, signature, or public — and implement it in the handler. Return 404 rather than 403 for resources whose existence is sensitive. If it handles a webhook, verify the signature over the **raw body**.

20.5 Modifying authorization

Change `src/server/services/context.ts` and nothing else. If a change makes a tenancy test fail, the change is wrong — do not adjust the test.

20.6 Required before merging

```
npm run typecheck
npm run lint
npm test
npx next build          # NOT `npm run build` — that migrates
npx playwright test
```

20.7 Things never to do

Never	Why
Run `npm run db:seed` against production	Creates `password123` accounts
Run `npm run build` locally without checking `DATABASE_URL`	Migrates whatever it points at
Edit an applied migration	Checksum mismatch breaks every environment
Weaken `assertTestDatabase()`	It is what stops E2E wiping a real database
Trust a client-supplied `studioId`	The tenant comes from the session
Catch `UserError` and swallow it silently	The planner needs the message
Add `NEXT_PUBLIC_` to anything secret	It is embedded in the browser bundle
Relax the CSP to make a script work	
Commit `.env*`	`.gitignore` covers `.env*` with `!.env.example`

21. External Integrations

Integration	Where	Failure behaviour
Vercel	Hosting, build, env injection	—
Neon	`DATABASE_URL` / `DIRECT_URL`; pooled vs direct endpoints	`/api/ready` reports `db: unreachable`, 503
Resend	`src/lib/email.ts`	Never throws; records `SKIPPED` / `FAILED`
Stripe	`src/lib/stripe.ts`, `subscriptions.ts`, `stripe-events.ts`, `/api/webhooks/stripe`	Publishing fails closed in production unless price is $0
Vercel Blob	`blobDriver()`	Falls back only if unconfigured, not if failing
S3-compatible	`s3Driver()`	ditto
Upstash Redis	`src/lib/ratelimit.ts`	Degrades to in-process limiter, logged

Not integrated: Sentry, Datadog, PostHog, Segment, Twilio, Cloudinary, Algolia, Remotion. *(verified against `package.json`.)*

22. Appendix: verification notes

22.1 What could not be verified

Area	Why
Runtime behaviour of Stripe subscriptions	Stripe has never been enabled in production; the subscription and webhook paths are covered by unit tests against mocks only
Actual production environment variable values	Not readable from the repository, and deliberately not documented
Vercel project settings (regions, protection, domains)	Configured in the Vercel dashboard, not in the repository — there is no `vercel.json`
Neon branch topology beyond names	Managed in the Neon console
Whether `ARCHIVED` weddings were ever used	The enum value exists; no code sets it

22.2 Documentation gaps

- **`loading.tsx` files** — none exist; there are no route-level loading states.
- **`MEMBER` role** — defined but unreachable. Its intended behaviour is undocumented.
- **`ARCHIVED` status** — defined but unset.
- **Help Center articles** — 26 exist in `src/lib/help.ts`; their content is not reproduced here.
- **CSS architecture** — `src/app/globals.css` is large and hand-written; this handbook covers the design tokens it exposes but not its full structure.

22.3 Commit provenance

This handbook documents **`f912143`**, the head of `origin/main`.

That commit added the Terms and Privacy acceptance system described throughout: `LegalAcceptance`, `src/lib/legal.ts`, `src/server/services/legal.ts`, `/terms`, `/privacy`, `/accept-terms`, the `requireStudio()` gate and migration `20260814120000_legal_acceptance`.

One later change is presentation only and does not affect anything documented here: the legal documents were moved off the marketing site's dark `.m-plate` onto a light, high-contrast document layout (`.legal` in `src/app/(marketing)/marketing.css`). The legal text, the version constants and the acceptance logic were not touched.

The commit immediately before it, `fe20a8b` ("Harden production launch"), differs by exactly five files — the invitation resend fix:

File	At `fe20a8b`
`src/server/services/invite-actions.ts`	absent
`src/components/InviteButtons.tsx`	absent
`tests/invite-actions.test.ts`	absent
`src/app/studio/weddings/[id]/guests/page.tsx`	resend action threw instead of returning an outcome
`tests/e2e/abuse.spec.ts`	waited on `networkidle` rather than the button state

Unit test counts at each point, so a mismatch tells you which state you are looking at:

State	Unit tests	E2E
`fe20a8b`	180 across 11 files	23
`eb056b3`	193 across 12 files	23
`f912143` (this document)	**214 across 13 files**	**29**

If you are ever reading this handbook against `fe20a8b`, 3.5, 9.3 and 16 describe behaviour that does not exist there.

22.4 Related documents

File	Contents
`README.md`	Quick start
`DEPLOYMENT.md`	Vercel, Neon, Resend, Stripe and Preview setup
`OPERATIONS.md`	Backups, monitoring, incident procedure
`SECURITY.md`	Security policy
`SECURITY_AUDIT.md`	Current verified security state
`SECURITY-AUDIT-2026-08.md`	The August 2026 audit
`EMAIL.md`	SPF/DKIM/DMARC and deliverability
`SUPPORT.md`	Support ticket system
`AUDIT.md`	Earlier audit history

End of handbook.